THREE TESS CANDIDATES

Further archival constraints

TIC 382098683 · TIC 188830138 · TIC 201247574
Completed analysis and provenance audit • 18 October 2026

All three remain unvalidated. Existing observations support repeatable transit-like signals, while surviving host and companion configurations prevent a unique planetary interpretation. This continuation retrieves spectra, corrects observation counting and RV provenance, tests physical blends, audits late events, and adds explicitly conditional primary-signal controls.

Candidate	Signal reliability	Host attribution	Planetary interpretation
382098683	Strong multi-sector coherence; new controls support the fixed signal.	Resolved-source localization retained. Unresolved companion hosts remain.	Two independent GALAH nights, not three RV epochs. APOGEE limits some separated secondary spectra; circular RV limits are conditional.
188830138	Strong coherence; new TGLC reduction retains the dip.	Named target favored among resolved stars. Faint blended hosts survive.	Current LAMOST error reduces Gaia discrepancy to 1.78σ. Circular bound twins disfavored; faint background twins still fit.
201247574	Four late events reproduced; baseline sensitivity gives S/N 4.09-5.13.	Prior image tests favor target; 13.7″ source needs nearly total eclipse.	Ordinary planet on that faint catalog neighbor cannot supply depth. Other unresolved/eccentric/diluted alternatives survive.

Material changes

The spectroscopy concern for TIC 188830138 is weaker than the report suggested. The third GALAH row for TIC 382098683 combines the other two observations and cannot be counted independently. TIC 201247574's fourth late event is only about half covered. Its neighbor's actual aperture contribution implies a nearly total eclipse, and its apparent timing shift is only about 2.3 formal σ after reconstructing the old ephemeris covariance.

No earlier conversational probability was used as a prior or target. No new validated planet, survey-wide false-alarm rate, or probability for "at least one of three" is reported. Repeated reductions and repeated analyses of the same observations are not independent confirmations.

1 Inputs and completed work

The actual local 26-page report was read in full, including methods, limitations, candidate pages and provenance. Its filename is **TESS_seven_candidate_continuation.pdf** and title is "SEVEN CANDIDATES / CONTINUATION AUDIT." Its candidate numbers, page locations and four late events match the request; the exact requested filename was not located. This identification is recorded, rather than assuming an unseen attachment.

The continuation and original-analysis directories contain 1,551 inventoried files totaling about 3.28 GB. The source light curves, event tables, pixel cubes, FPP scenario tables, archived scripts, receipts and package files were actually available. The original discovery repository and raw light-curve store were also found. A reference in the PDF was never treated as proof that a file existed.

Workstream	Status	What was done / remaining limit
Report and source bundle	Completed	Full report read; actual inputs inspected; inventory and SHA 256 provenance retained.
Ephemerides	Completed with limit	True packet epochs and marginal errors recovered. Original joint covariance absent; conditional timing likelihoods reconstructed.
Previous PRF / secondary / splits	Retained	Substantial completed tests not rerun for extra apparent evidence. New throughput and S28 tests address specific gaps.
Spectroscopy	Completed, partly inconclusive	Full GALAH/APOGEE/LAMOST products and visits retrieved; deduplicated; conditional RV and line sensitivity tested. No complete companion census.
Physical alternatives	Completed finite maps	Native-integrated twin models, secondary-host examples, RV/shape and flux-budget constraints. No full identifiable evidence integral.
Archives / reductions	Completed bounded search	Current documented routes, valid bulk catalogs, three TGLC files, selected original FITS. Some target routes fail; no phase-covered ground data for 201 found.
Primary reliability	Conditional tests complete	Fixed-phase controls, matched stars and processing injections. Full original survey search/selection calibration remains missing.
Validation / mass	Inconclusive	No new mass measurement or calibrated contrast curve resolves all remaining hosts.

Source report SHA 256: 07 adf 2 dc 775 db 478 ec 42997 bc 6 ecdc 0 ec 9 ec 6260463185021c08e5 ccc 0f5b33b. The input inventory and manifests identify exact source paths; original analysis files were treated as read-only.

2 Verified identities and ephemerides

TIC	Gaia DR3 / 2MASS	Gaia 2016 coordinates (deg)	Proper motion (mas/yr)
382098683	4762639241115869568 05195347-5940587	79.97273635, -59.68267333	-8.787, +71.169
188830138	1416467672916352768 17340638+5205471	263.52674314, +52.09627762	+13.545, -36.779
201247574	4918283671897810560 00005586-5910378	0.23361442, -59.17720622	+97.624, +3.679

Identifiers are stored as strings to avoid rounding Gaia IDs. Coordinate searches either propagated the proper motion to the observation date or used a conservative motion-expanded tolerance, followed by identifier checks. The LAMOST fiber position is 0.705″ from the propagated target.

TIC	Frozen period P (days)	Frozen T0 (BTJD_TDB)	T14 (hours)
382098683	4.351684565480 σP = 0.00000753272	1329.1390161814 σT0 = 0.00329561 d	1.738 ± 0.178
188830138	1.181024043263 σP = 0.00000303505	1764.8908894882 σT0 = 0.00345149 d	1.640 ± 0.172
201247574	16.560576472191 σP = 0.00005222531	1333.6099055691 σT0 = 0.00632544 d	3.240 ± 0.326

BTJD_TDB = BJD_TDB - 2457000. These are recovered packet values, not invented epochs. For cycle n, t(n)=T0+nP and Var[t(n)]=C00+2 nC 01+n^2C11. The original joint fit covariance was not saved; combining only the marginal errors as if uncorrelated is not justified.

TIC	Reconstructed T0 / P (days)	C(T0,T0), C(T0,P), C(P,P), days^2
382098683	1329.1366007693 4.351686267024	8.51359e-6, -1.82551e-8, 5.24337e-11
188830138	1764.8885043702 1.181025727489	1.23841e-5, -1.05929e-8, 1.09438e-11
201247574	1333.6071568158 16.560611718754	4.16821e-5, -3.20715e-7, 3.15076e-9

These reconstructed covariances come from local-baseline trapezoid timing likelihoods with residual-noise treatment. The first two hold duration fixed; the 201 fit frees depth, duration, epoch and period using ten old events only. Full covariance arrays are supplied. They are conditional reconstructions, not recovered original posteriors or recommended definitive ephemerides. Frozen predictions were retained for the registered tests.

3 Archive census and dispositions

Canonical ExoFOP redirects to /tess/help.php. Its documented PHP/export instructions were retrieved. Per-target HTML/JSON and upload/imaging/spectroscopy/time-series routes returned malformed "Array" content, even with HTTP 200. Those requests are failures, not empty scientific results.

Archive / route	Actual outcome	Interpretation
ExoFOP prebuilt exports	36,971 imaging; 34,775 spectroscopy; 22,434 time-series rows. Zero exact-TIC entries for all 3.	Valid parsed exports. No entry under these TIC IDs; alternate aliases, unpublished observations or missing uploads remain possible.
Current TOI / CTOI	8,151 / 5,138 rows; zero identity matches with TIC and motion-aware coordinate safeguards.	No matched TOI/CTOI disposition to confirm by epoch. This is not a validation verdict.
NASA PS / PSCompPars	Live TAP failed with Oracle connection error. Cached 15 Oct tables:39,502 /6,144 rows, hashes verified; zero identities.	Cached absence explicitly dated. The cached NASA extract lacks ephemeris columns.
SPOC TCE snapshot	344,404 cached rows across 134 files; zero identity matches.	No candidate-matched mission DV disposition found. No mission 2-minute light curve identified in current target query.
TESS-EB	4,584 current rows. No target identity matches. Broad 5° alias search checked.	One 382 near 5:2 alias at 4.25° fails event-train matching; no supported contamination match.
High-resolution imaging / follow-up	No matched calibrated contrast curve or phase-covered follow-up retrieved.	No imaging non-detection or contrast exclusion is invented.
201 ground images	SkyMapper 221 metadata rows, none at useful primary phase; DES valid empty; NSC timeouts.	Different outcomes retained separately. No forced photometry without accessible useful images.

The EB alias TIC 350297040 has P=10.8739196 d; 5P382-2 PEB=0.010584 d. Only 9/227 admitted 382 centers overlap either catalog eclipse, or 13/227 allowing conservative timing uncertainty (wider Gaussian widths:12/227 or 17/227). It cannot explain the recurring train by a simple ephemeris match. The bounded 5° search is not an exhaustive detector-plane ghost/rolling-band census.

MAST target/footprint searches identify TESS-SPOC, QLP and applicable TGLC, TASOC, TARS, T16 and GSFC-ELEANOR-LITE products. Counts and IDs are archived; footprint hits are not automatically target light curves. Alternative labels sharing pixels or extraction methods do not supply independent observations.

4 Photometry, timing and dilution

The actual input product trace matters more than a column label in a downstream file. TIC 188 uses 19 photometric sectors, while the report's20-sector statement concerns image/localization coverage. TIC 382 uses 42 sectors. TIC 201's original fitted light curve uses 7 sectors; its four later events come from independent TESSCut extractions in 102 and 105, already in the report.

Candidate	Actual QLP sectors in saved fitted input	Timing issue exposure
382098683	87,88,89,90,93,94,95,96,97,98	No QLP 74-79 or 99-104 files used.
188830138	None; all selected products TESS-SPOC	No affected QLP file used.
201247574	95,96; other old sectors TESS-SPOC	S102/105 late events are TESSCut, not QLP.

Current MAST notes identify erroneous QLP barycentric corrections in sectors 99-104, including a different ephemeris-column error beginning in 102; the note does not establish corrected replacements for the affected files. Sectors 74-79 were replaced withv 02 products after an earlier reference-plane error. Neither affected QLP set entered these saved analyses, so no used file needed replacement. QLP began using the TGLC extraction method in sector 94. [1,2]

Representative re-fetched TESS-SPOC files 382/S31,188/S53,201/S28 match the original recorded FITS hashes exactly. Stored raw flux and times match PDCSAP_FLUX at ingested timestamps. CROWDSAP is 0.98223,0.96906,0.97057 and FLFRCSAP 0.81825,0.84290,0.80379 respectively. PDCSAP is corrected for estimated aperture losses and resolved crowding. Normalization and biweight detrending introduce no additional crowding division.

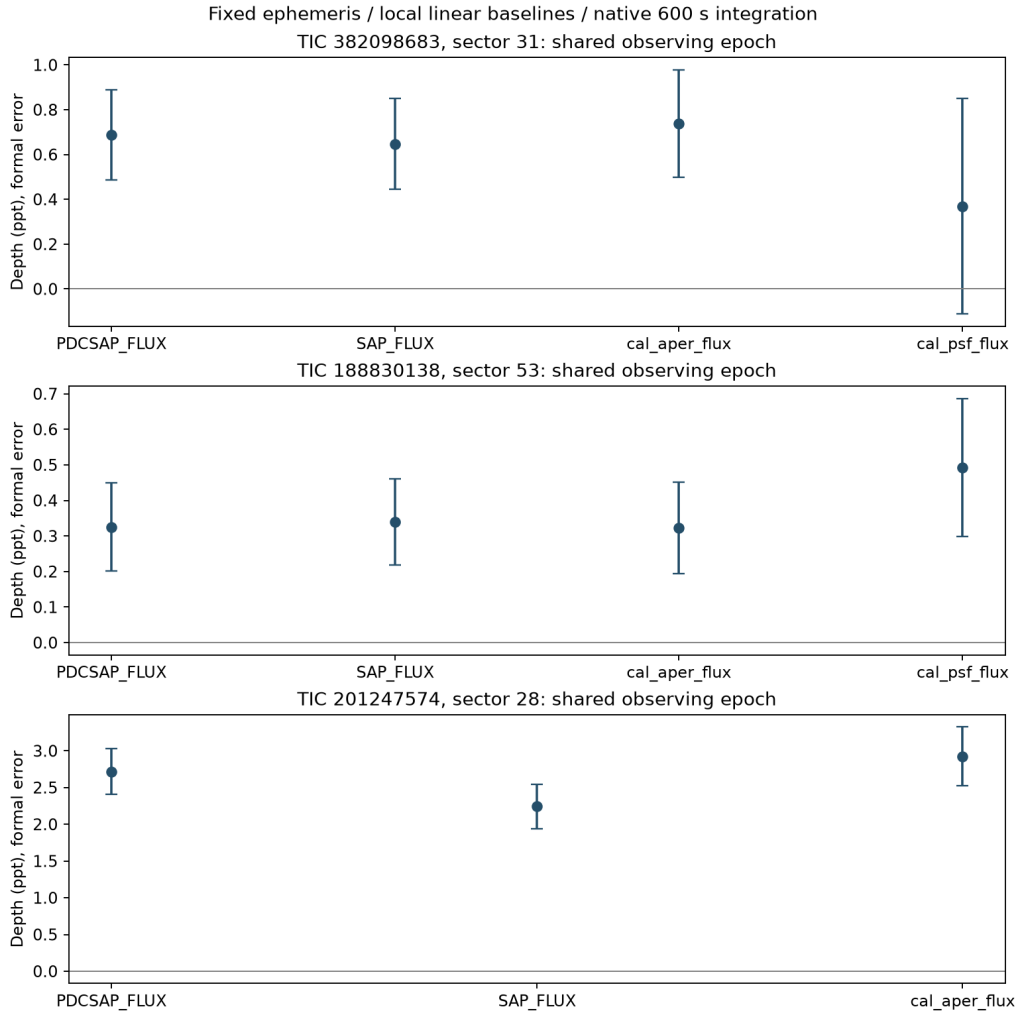
QLP DET_FLUX replaced KSPSAP_FLUX as the detrended column; SYS_RM_FLUX removes systematics while retaining stellar variability. QLP uses catalog-referenced photometry and, from 94, Gaia/ePSF decontamination. This supports the report's corrected-target convention, but neither catalog photometry nor a scalar scene fraction proves exact sector-by-sector deblending. The retrieved TGLC files supply already decontaminated, normalized calibrated aperture/PSF fluxes. No raw TESSCut depth was silently treated as dilution-corrected.

TRICERATOPS dilution_corrected=True leaves the named-target input unchanged and transforms nearby-host flux with the target/neighbor scene ratio. Hypothetical unresolved companion dilution remains part of each physical model. No extra target-depth division was found. Cached scene fractions are not exact actual-aperture throughputs in every sector. White folded-bin errors and fixed stellar parameters remain material limitations.

Native sampling is integrated at 200,600 or 1800 s as appropriate. TGLC's XPTIME=600 s is used, rather than its elapsed-span/row-count TIMEDEL. TESSCut uses 200 s cadence span despite 158.4 s accumulated exposure after deadtime. Its inherited CCD-center barycentric correction differs by only about 24 s at the late 201 events, far below the apparent 25 min timing offset. [3]

5 New reduction checks on existing epochs

MAST explicitly documents that TGLC sectors 12-55 can be absent from Portal/API indexing. The sector 31,53 and 28 target lists were streamed until the exact Gaia/TIC identity was found; only the needed FITS files were downloaded. These are new reductions inspected here, not new observing epochs. [1]



Fixed frozen ephemeris, native 600 s integration and local linear baselines. Error bars are formal and different reductions share photons. All signed measurements are retained. TIC 201 TGLC PSF flux is entirely nonfinite and NEAREDGE is true; only its aperture series can be measured.

TGLC aperture depths:382/S31 0.738±0.240 ppt;188/S53 0.323±0.129 ppt;201/S28 2.923±0.402 ppt. Corresponding simple PDCSAP estimates are 0.687±0.201,0.325±0.124 and 2.719±0.315 ppt. No significant independent confirmation statistic is formed from these comparisons.

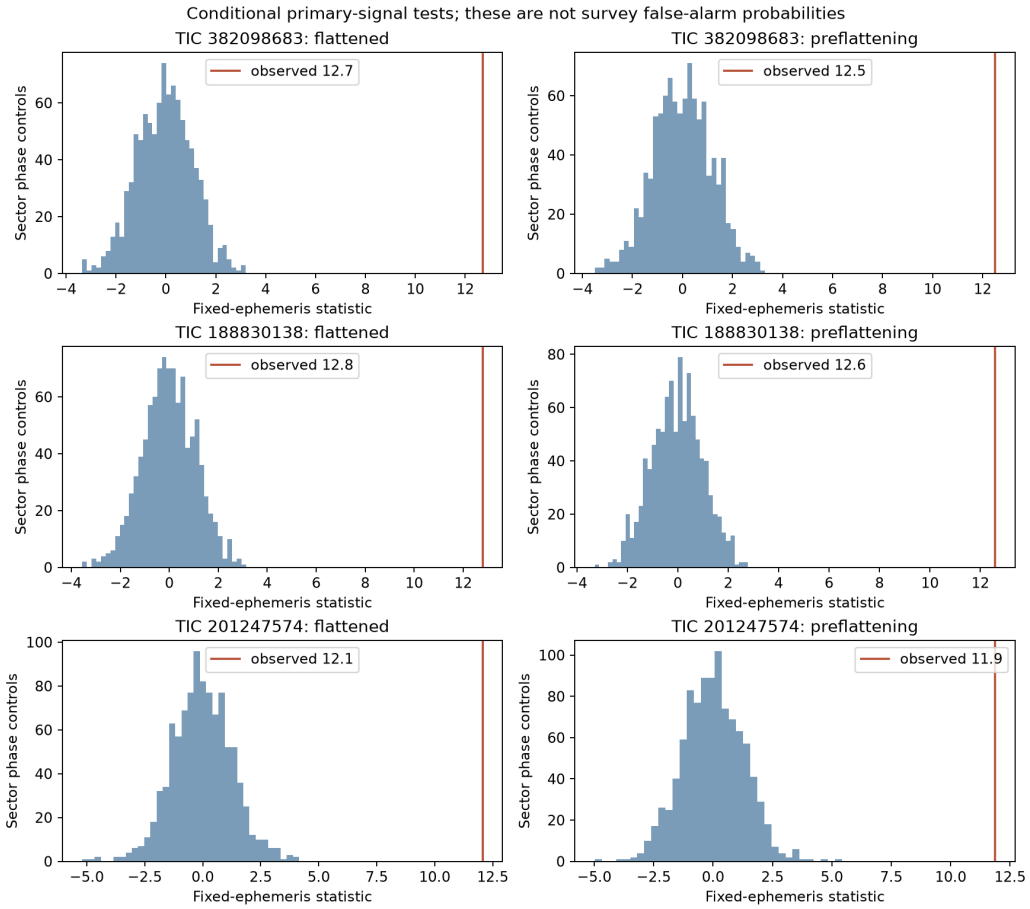
The higher 201/S28 depth triggered a bounded follow-up. Both primary events have 20/20 good in-event samples; it is not a partial primary. With per-event residual-noise inflation, PDCSAP changes from 2.588±0.347 ppt for linear baselines to 1.959±0.310 ppt for quadratic baselines. Quadratic SAP/CROWDSAP≈1.944 ppt is consistent. TGLC gives 2.74-3.13 ppt across these choices, has no usable PSF series and no pixel cube. Baseline/reduction sensitivity prevents interpreting this difference as astrophysical depth variation.

Exact filenames follow hlsp_tglc_tess_ffi_gaiaid-[GaiaID]-s0031-cam4-ccd1, -s0053-cam 2-ccd 2 and-s0028-cam 2-ccd 2, with suffix_tess_v1_llc.fits. Full product names, URLs, headers, quality selections, integration and SHA 256 hashes are in reduction_comparison.csv and archive receipts.

6 Conditional primary-signal reliability

A protocol was saved before the primary controls:1,000 fixed-seed independent sector phase offsets, frozen P/T0/T14, signed finite-integrated trapezoid statistic, local linear baselines, coverage admission by point counts, and no selection by fitted depth. Offsets avoid the known transit; real timestamps, gaps, flux sequences and sector structure remain intact. Both saved flattened flux and reconstructed pre-flattening flux were tested. Earlier clipping cannot be undone by multiplying the trend back in.

TIC	Flattened depth ± formal σ (ppt)	Fixed statistic	Pre-flattening statistic	Control exceedances
382098683	0.4318 ±0.0340	12.70	12.50	0/1000 in each
188830138	0.3478 ±0.0272	12.79	12.59	0/1000 in each
201247574	1.5856 ±0.1314	12.06	11.87	0/1000 in each



The observed fixed-ephemeris statistics exceed the registered 99th-percentile criterion. Sector phase null standard deviations range 1.005-1.294, showing departures from ideal unit white noise. This figure does not calibrate the original discovery search.

For each zero-exceedance test, the add-one estimate is 1/1001=0.000999; the nominal independent-binomial two-sided 95% upper bound is 0.00368. Controls share data and are dependent; that interval does not represent all noise-model uncertainty. Six tests are sensitivity checks, not six independent confirmations. Zero observed false alarms is not zero risk.

7 Controls, injections and the discovery gap

The original discovery code, configuration, sample and history were located. The search spans 0.5-50 d, durations 0.02-0.35 d, a40,000-point grid plus local refinement, up to 6 signals and whole/chunk searches. An 8,100 s target wall-clock cap and adaptive development/selection history complicate exact replay. The S1 sample contains 10,000 targets. A full same-search, same-final-selection ensemble is not executed or represented as complete here.

Existing K 1 controls used 250 reserved stars, which are not verified pure noise:17.376 accepted search signals/star,2.692 pre-gate/star, and 0.088 A/S bound-alias passes/star. Those counts are not the final pipeline's individual false-alarm rate. K 2's70 fresh adversarial EB/blend tests produced 0 SUBMIT and 1 HOLD, with an upper Wilson scale about 5.2%; these test selected astrophysical negatives, not general instrumental discovery reliability. No published LEO-Vetter reliability is transferred to these targets. [8]

Matched stars processed with the original preprocessing

Target	Controls completed	Control statistic range	Scope
382098683	8/8	-0.70 to 1.44	Fixed candidate ephemeris; no blind search
188830138	8/8	-0.80 to 1.27	Fixed candidate ephemeris; no blind search
201247574	8/8	-0.99 to 0.66	Fixed candidate ephemeris; no blind search

Eight available survey stars per target were selected before consulting flux: within 6°, ranked by T magnitude, temperature, sector count and separation. Original quality selection,1.1 d biweight and clipping were run on their native stored products. The sample is small, not guaranteed astrophysically blank, and only tests whether these stars reproduce the candidate's fixed ephemeris. Even 0/8 exceedances has a nominal 95% upper tail bound 0.369; it is not a useful discovery-rate estimate.

Signals injected before the processing being checked

A second fixed 24-case test injected finite 600 s transits into both corrected and SAP native flux before the actual original quality/biweight/clipping steps:8 offsets each in 382/S31,188/S53 and 201/S28. There are 22 coverage-admitted cases. Most incremental depth responses are approximately 0.998-1.015 of injection, but one 382/S31 case gives 0.492 as the cadence clipping selection changes by one sample. That failure is retained. The two 201 cases with no admitted event are coverage failures, not non-detections. This finite test is not an efficiency curve or proof that discovery clipping is harmless.

For the four late 201 events, a separate predeclared 1,000-control test has 0 exceedances;64 injections were added to raw pixels before sky extraction. The 183 admitted injected events recover 1.541-1.556 ppt for a1.56 ppt intrinsic target decrement, consistent with throughput. These controls, injection tests, previous secondary tests and PRF tests are not multiplied into a single reliability posterior.

8 TIC 382098683: spectra and RV provenance

Gaia DR 3 4762639241115869568 /2 MASS 05195347-5940587 is securely identified. Full optical and H-band products were obtained, beyond the original narrow red-arm wrapper. Two original "FITS" downloads are actually empty VOTables. The usable original SSA wrapper has no error spectrum; its RV header is the barycentric correction, not the stellar RV. It cannot support a precision or secondary-line exclusion by itself.

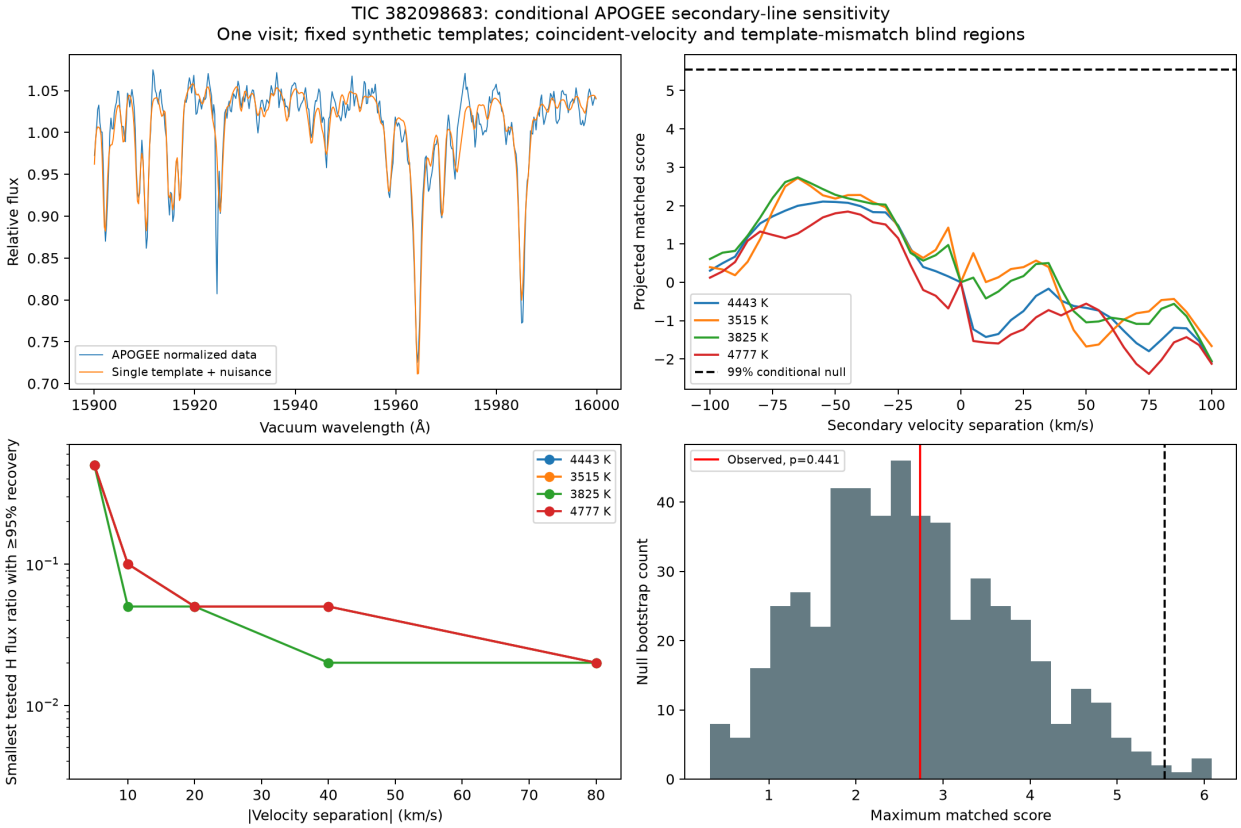
Observation / product	Independent epoch	RV / convention and provenance
GALAH 131120003001222	2013-11-20; startMJD 56616.6537034	DR 3 no-GR RV-5.627±0.075 km/s; gravity-corrected-6.237±0.111. Midpoint not recovered; +0-2 h bracket used.
GALAH 131216001101079	2013-12-16; midpointMJD 56642.5749479	DR 3 no-GR-5.832±0.059; gravity-corrected-6.453±0.109. DR 4-5.894±0.189 is a reprocessing of this epoch.
GALAH 131216001102079	Combined product; no third epoch	Headers COMB_2_0/1 identify the two observations above. Its RV must not be an independent likelihood row.
APOGEE apogee.lco 25m.dr 17.11188.58774.141	2019-10-18; JD 2458774.73733	One visit, barycentric RV-6.24815±0.04998 km/s; raw-8.322 is not comparable. asVisit/asStar/aspcapStar retrieved.

GALAH original four-arm science/error data and the DR 4 December spectrum were retrieved. The APOGEE visit includes flux, errors, masks and CCF information; STARFLAG=0 and RV_FLAG=0, one reported CCF component. Its 8 frames total 4004.08 s; EXPTIME 2002.04 s is per dither position. ASPCAPFLAG=16388 flags microturbulence and rotation warnings. Product versions, S/N, spectral ranges and request IDs are in the inventory. [4,5]

The earlier three-row constant-RV p≈0.035 is not valid. With two consistent gravity-corrected GALAH measurements, ΔRV=-0.216±0.156 km/s and p=0.165. The same-star no-GR pair is more directly comparable:-0.205±0.095 km/s,2.15 formal σ, p=0.0317. Adding a0.1 km/s per-epoch floor gives p=0.230; a0.3 floor gives p=0.637. Deduplication does not erase every formal hint, but no calibrated variability or orbit is established.

Survey zero points, template/gravitational conventions and activity are not averaged down as three independent precise measurements. The joint RV calculations use the no-GR pair and the single APOGEE visit, with explicit error floors and a nuisance survey offset. Gaia mean velocities are not assigned artificial observing times.

9 TIC 382098683: secondary-line sensitivity



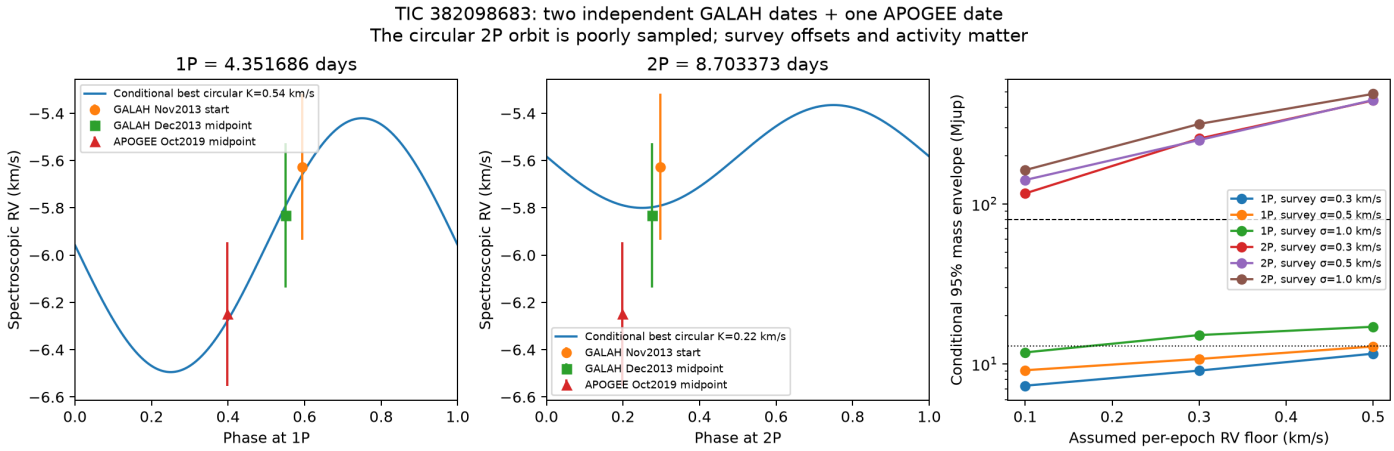
APOGEE composite-template sensitivity with additional template/detector centroid nuisance freedom. The test is conditional on the four template families, flux-ratio convention, velocity grid and residual noise model; it is not a universal companion exclusion.

The analysis uses 6,714 selected spectral pixels, synthetic families near 3515,3825,4443 and 4777 K,512 block-wild null realizations and 64 injections per grid point. Companions are injected into composite spectra before nuisance refitting. H-band secondary/primary flux ratio qH is not the secondary fraction of total light (that fraction isqH/(1+qH)).

Separation / flux ratio	Measured recovery	Interpretation
|Δv|20 or 40 km/s; qH=0.05	64/64 for all 4 tested families; Wilson 95% lower 0.943	Strong conditional sensitivity to separated companions of these types.
|Δv|80 km/s; qH=0.02	62-63/64 across tested families	Some fainter well-separated companions detectable.
Δv≈5 km/s	qH≈0.5 needed for≥95% in tested grid	Severe blending blind region.
Δv=0 or untested cool/broadened types	No universal exclusion	Near-identical velocities, very cool/faint stars, rotation and metallicity mismatch remain.

A +5 km/s feature in the simpler residual model was not robust. With extended template/detector centroid nuisance terms, observed score 2.736 is below the 99% null threshold 5.541 (conditional tail 0.441). It is not claimed as a detected companion. The nuisance expansion was a post-detection robustness check, not another independent search. These data do not exclude every spectrally single-looking unresolved binary.

10 TIC 382098683: physical constraints



Actual deduplicated phase sampling and conditional RV constraints. Circular and eccentric configurations must be kept separate; the survey offset and realistic noise floor materially affect the limit.

For a circular named-target orbit atP=4.3516846 d, a0.3 km/s per-epoch floor and 0.5 km/s survey-offset prior give a conditional K 95 envelope about 1.68 km/s, corresponding to 9.49-10.75 Mjup forMstar 0.58-0.70 Msun. The grid over floors 0.1-0.5 and offset widths 0.3-1.0 km/s spans 6.46-17.07 Mjup. These are conditional profile limits, not an eccentricity-marginalized mass posterior.

At 2P, the three actual epochs cluster at phases 0.199-0.297. The same middle assumptions permit K≈25.8 km/s, or roughly 224-251 Mjup. Therefore those RVs cannot broadly exclude twice-period stellar/brown-dwarf explanations. Selected eccentric orientations also evade much tighter circular limits.

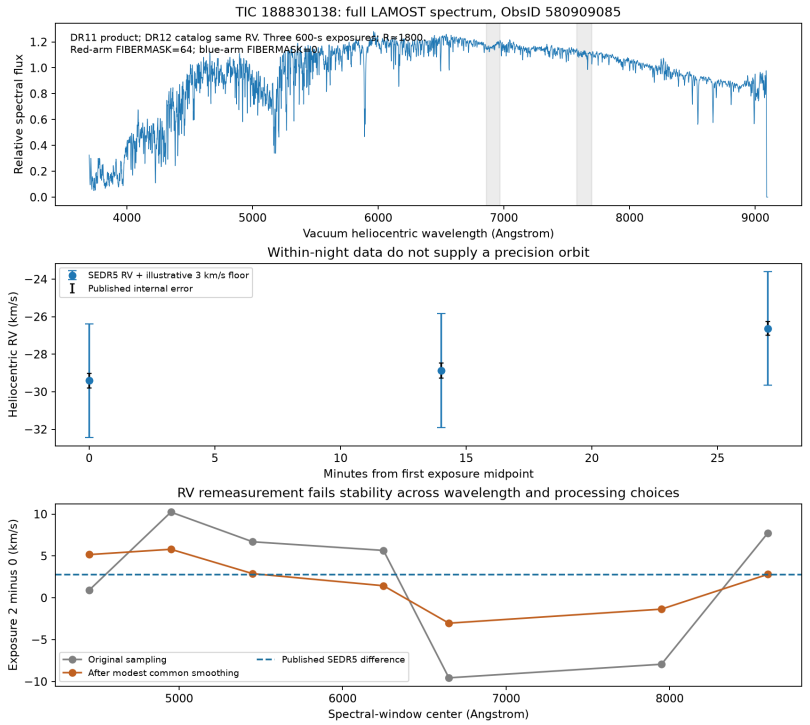
A bounded joint shape/RV calculation includes companion mass in Kepler dynamics, finite cadence, Mstar 0.58-0.70 and Rstar 0.58-0.68. A13 Mjup,0.10 Rsun grazing example fits the RVs at e≈0.897 andω≈-94.7°, while paying joint $\Delta\chi^2$15.79-16.18. The 80 Mjup and 150 Mjup examples pay 24.89-40.61 and 46.68-93.27 over tested errors; all best fits reach stellar-property boundaries. Small departures from exact apastron change the RV penalty substantially. These selected, fine-tuned solutions do not establish equal probability or an entire mass-class exclusion.

DR 3 GALAH and APOGEE indicate metallicity near-0.30; DR 4 is lower, about-0.45 to-0.54. Catalog formal errors cannot be combined as independent constraints. Parallax≈7.814 mas and the SED constrain total luminosity. The earlier free-normalization single-star fit versus fixed mean-sequence binary fit is not a matched binary-evidence comparison. The new linked 0.15-0.30 Msun host map requires about 4.3-3.1 Rearth occultors, minimum central-transit eccentricities≈0.32-0.12 and qH≈0.030-0.130, with contrast sensitivity included. Many hosts are cooler than the lowest injection template or can have coincident velocities. No metallicity-marginalized binary posterior or numeric STP reduction is justified.

11 TIC 188830138: current spectroscopy

The 2017-06-14 LAMOST observation is secure: ObsID 580909085, J173406.39+520547.2, plateHD 173133N504043V01, spectrograph 09/fiber 085. DR 11 and currentDR 12v2 documented 5″ cone searches each return one LRS observation and zero MRS rows. DR 5/DR 11/DR 12 are repeated reductions of the same night. Full coadds and all three 600 s blue/red exposures were retrieved. [6]

Measurement	RV (km/s)	Meaning
LAMOST DR 5 report claim	-30.08±3.02	Earlier formal error; faithfully quoted in prior report.
LAMOST DR 12v2	-30.07±5.52	Same observation; revised uncertainty.
Gaia DR 3 median	-20.146954±0.689995	15 transits,13 visibility periods over 845.35 d. No assigned epoch.
Current discrepancy	9.923 /5.563 =1.784σ	Not an established orbit or strong cross-survey RV conflict.



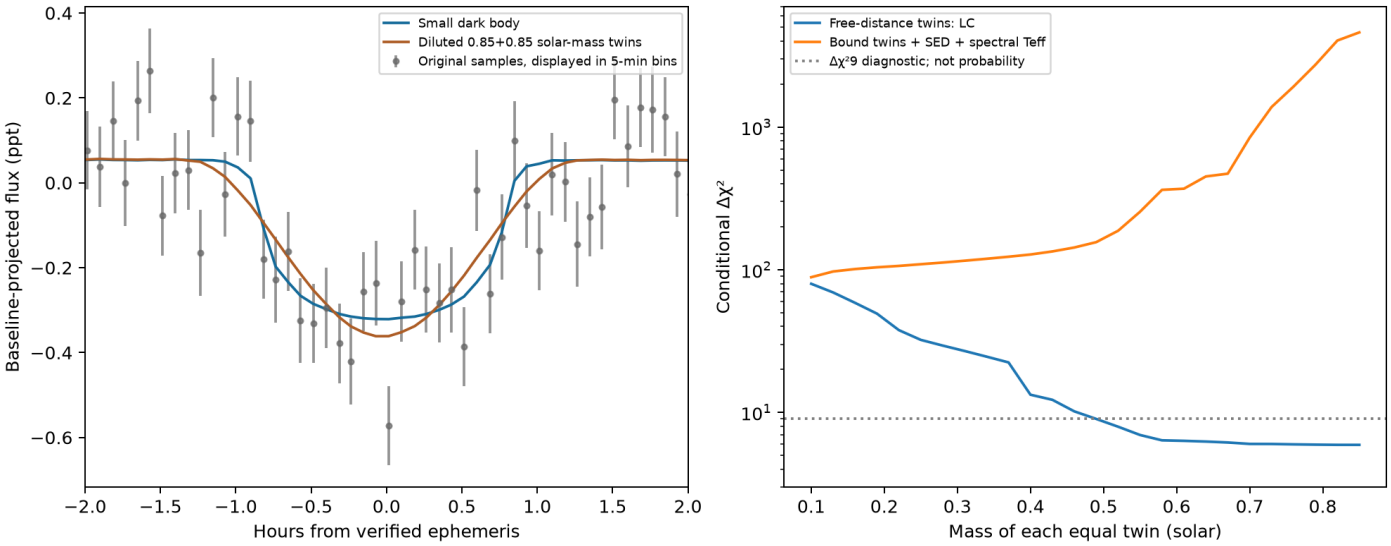
Full-spectrum and single-exposure diagnostics. The current uncertainty and systematic sensitivity matter more than a comparison using old formal errors alone.

Individual midpoint BJD_TDB values are 2457919.1536618,2457919.1633840 and 2457919.1724118 (minute-level start-time quantization). Catalog RVs-29.41±0.39,-28.87±0.39,-26.63±0.36 span 27 min, not separate nights. Formal-only constant-RV p=1.4e-7 becomes 0.148 or 0.788 with 1 or 3 km/s sensitivity floors. Relative remeasurement changes by several km/s across wavelength windows and smoothing; no stable new absolute RV is claimed.

The vacuum spectrum is already heliocentric-corrected. FIB_MASK64 is the WHOPPER bright-fiber relation flag, not a clean zero flag; the reported neighboring-fiber ratio is 0.02. Coadd S/N is high, but R≈1800 cannot quantify faint near-coincident companion exclusion. Gaia's aggregate variability p≈0.660 is not an orbital limit without sampling. A wider companion could cause RV variability while a planet produces the dips.

12 TIC 188830138: physically linked twins

Native 36,693 samples across 336 coverage-admitted events were fitted with per-event baselines and 30 min residual-noise inflation. All depth signs were retained.200/600/1800 s integration was included. Stellar masses, radii, colors, luminosities, parallax, SED and a150 K spectral-temperature floor were linked; the same SED was not multiplied twice. Nearly solar LAMOST metallicity supports the adopted approximate main-sequence relation, but age/metallicity are not fully marginalized.

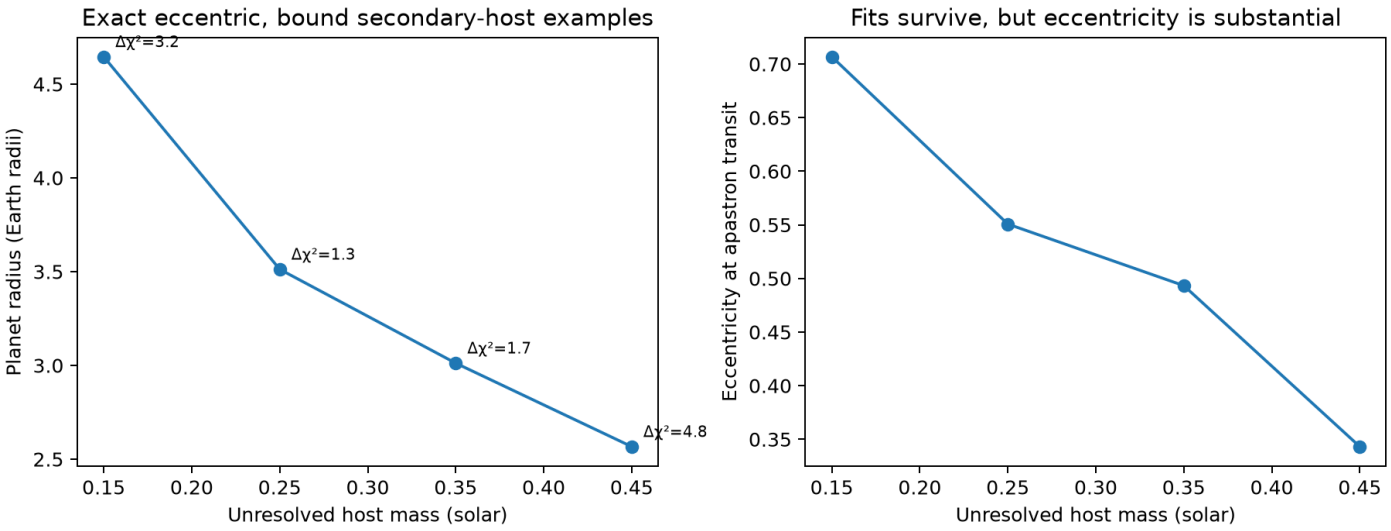


Exact equal-depth eclipses at 2.3620480865 d can resemble the transit. Bound triples and free-distance diluted twins occupy very different luminosity constraints. $\Delta\chi^2$ is a conditional fit diagnostic, not a posterior probability.

Configuration	Numerical result	Effect
Circular bound main-sequence equal twins;0.10-0.85 Msun each	Minimum combined $\Delta\chi^2$88.67; radius sensitivity minimum 83.46	Strongly incompatible within this finite ordinary-star grid. Not a universal triple/EB exclusion.
Free-distance equal twins;0.49-0.85 Msun each	$\Delta\chi^2$5.93-9.01; total flux fraction 0.00138-0.00383	Surviving faint background blend examples. Grid edge is not a mass upper limit.
Representative 0.85+0.85 Msun twins	R 0.7935 Rsun each; a/R 11.226; b1.311; ΔT6.04 mag; K 94.75 km/s	Deep intrinsic eclipses hidden in<0.4% total light. The visible target need not have this RV amplitude.
Diluted ellipsoidal signal	0.0006-0.004 ppt (α=1.5)	Below old 0.0491 ppt harmonic bound; undiluted modulation arguments cannot reject these models.

Equal-depth twins necessarily satisfy odd/even agreement. Excluding selected circular undiluted geometries does not exclude these diluted models. Spectroscopy of the bright target alone can miss their<0.4% light contribution.

13 TIC 188830138: secondary-host planets



Linked unresolved secondary-host examples and their finite-cadence transit fits. Eccentricity, luminosity dilution and physical radius are linked; fit quality is not an occurrence probability.

Secondary mass (Msun)	RP-band flux fraction	Occultor radius (Rearth)	e at ω=270°	Total $\Delta\chi^2$	Fluid Roche screening
0.25	0.0243	3.51	0.551	1.30	Mplanet≥17.1 Mearth
0.35	0.0486	3.01	0.493	1.74	Mplanet≥7.53 Mearth
0.45	0.1150	2.57	0.343	4.82	Mplanet≥2.14 Mearth
0.15 (extreme)	0.00577	4.65	0.707	3.19	Density≥7.81 g/cm^3;≈143 Mearth

The 0.25-0.45 Msun examples avoid direct collision and can pass the stated periastron Roche screening with plausible dense Neptune/sub-Neptune masses. The 0.15 Msun example requires an unusually dense 4.65 Rearth body; it is retained as an extreme shape example, not promoted as an ordinary plausible Neptune. The Roche coefficient 2.44 is a fluid-equilibrium diagnostic applied at periastron, not a tidal-evolution calculation.

These short-period eccentric configurations deserve strong eccentricity and tidal-survival prior penalties. The available data do not determine those priors or their integrated evidence. Circular planets on the sampled faint hosts fit poorly, but the entire unresolved-host class is not removed. No best-fit $\Delta\chi^2$ has been converted into an FPP.

The retrieved LAMOST products cannot identify a precision wavelength/LSF correction or reliably calibrate faint secondary-line completeness. A validated empirical-template recalibration could investigate the within-night drift, but more scalar fit refinement cannot tell which surviving photometric host is real. A wider stellar companion plus a target-host planet explicitly remains viable.

14 TIC 201247574: all four late events

These are the report's existing sector 102/105 observations, not new independent evidence. The exact extraction was reproduced, then audited for coverage, background, aperture, flags and timing. Event admission depended on coverage, not depth or sign. Sector 103's inadequate baseline coverage is not a non-detection.

Sector / cycle	Frozen center BTJD_TDB	Depth ± error (ppt)	Selected / raw in-event
102 /169	4132.347329369455	1.15361±0.38241	58/58; flags 0
102 /170	4148.907905841646	1.29091±0.37497	58/58; flags 0
105 /174	4215.150211730411	1.86645±0.39618	58/58; flags 0
105 /175	4231.710788202602	2.98408±0.71718	30/59;29 flag 128 excluded

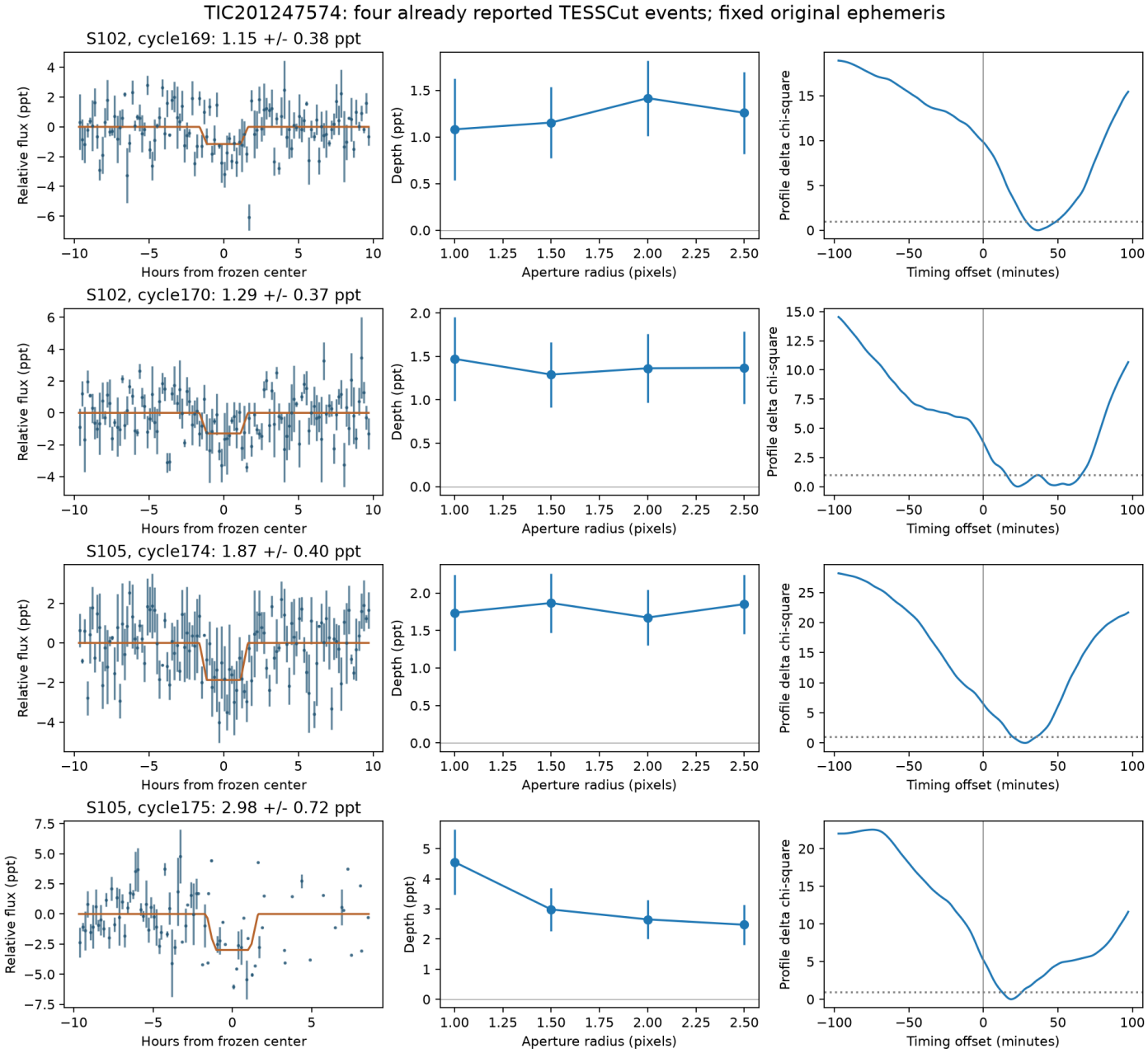
Pooled reproduction: **1.561268±0.304304 ppt, S/N 5.1306**. Depth scatter χ^2=6.19 for 3 dof, p=0.103, does not establish variability. Cycle 175 has 51.4% nominal coverage and a good-data gap edge 2.90 min from center; its higher depth is more sensitive to partial coverage. It remains in the primary result. Removing it is only a diagnostic, yielding 1.425±0.222 ppt; this is not used as a selected improved significance.

Strict QUALITY=0 gives the same four event measurements. The original bitmask retained some stray-light flags elsewhere; that affects sector 103's coverage interpretation, not these four events. Products are real TESSCut/SPOC FFIs, Astrocut 0.11.1, releases 140/141/143. S102 camera 4/CCD 3 and S105 camera 2/CCD 1 have different detector positions. Exact FFI names and source hashes appear in the event manifest.

The raw aperture flux subtracts a fitted sky plane and normalizes by its sector median. No target-dilution correction was applied. Background-template coefficients at the two 102 events have opposite signs; they are not calibrated eclipse significances. Quadratic and sky-covariate baselines preserve positive depths, but pooled S/N ranges 4.09-5.13 and depth 1.46-1.57 ppt. A single unqualified 5σ statement hides this sensitivity.

The independent event timing profiles prefer 18-37 min later centers. A shared offset fit gives+25.07±5.03 min, with similar quadratic result. Old-only covariance predicts centers 4.62-4.92 min later than frozen, with 6.95-7.33 min errors. The difference is only≈2.3 formal σ. Shape, correlated noise and reduction uncertainty are not fully marginalized; no TTV or definitive revised period is established.

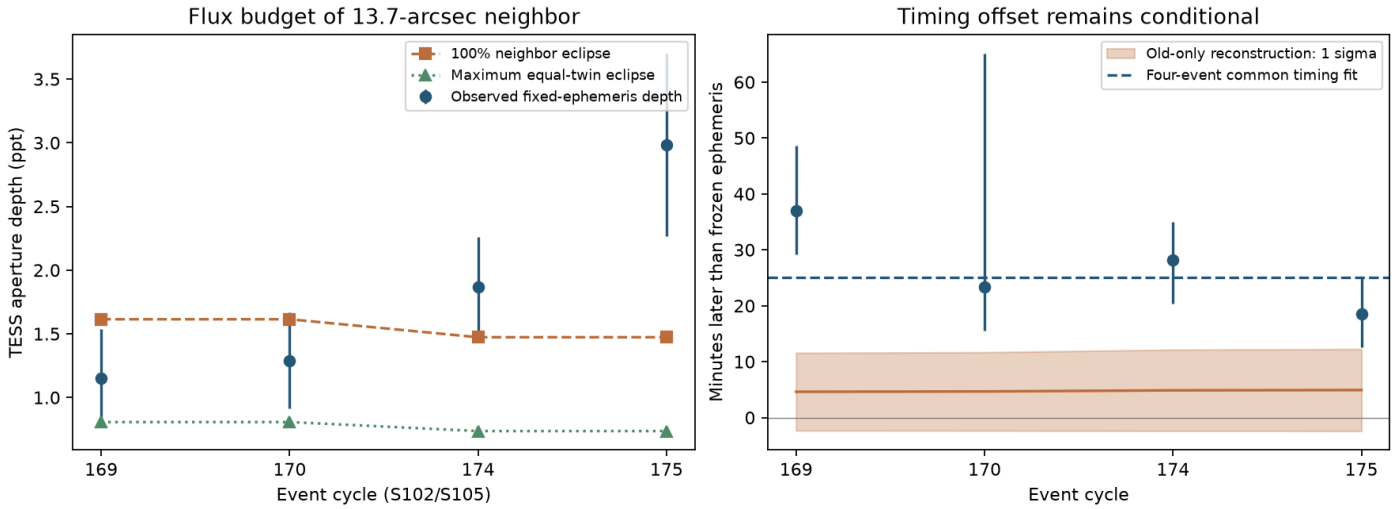
15 TIC 201247574: event diagnostics



All four events, aperture sensitivity and timing profiles. The incomplete last event is retained. Apertures share pixels; their agreement cannot be multiplied as independent evidence.

Raw-pixel injections cover extraction sensitivity:64 predeclared phases produced 183 coverage-admitted event measurements. A1.56 ppt intrinsic target decrement yields 1.541-1.556 ppt recovered aperture depth. This checks the sky and aperture pipeline, not all false positives. The conditional late-event phase controls preserve sector spacing and require four admitted events:1,000 accepted draws out of 4,307 attempts, zero exceedances. Its dependence caveat and distinction from discovery FAP are the same as the main controls.

16 TIC 201247574: which star can vary?



The actual-neighbor flux budget and the old-only covariance envelope. A nearly total faint-neighbor eclipse survives; this is not a calibrated host probability.

The 13.672″ source is Gaia DR 3 4918283667602203520 /TIC 616030855. Parallax differs by 24.95 formal σ and proper-motion vectors by 111.13 mas/yr, so it is not an ordinary bound companion. At catalogT=18.904 and the measured mission-PRF throughputs, it contributes 0.001614 and 0.001473 of the adopted S102/S105 aperture light (RP proxy 0.001542/0.001407).

A shared intrinsic eclipse requires **0.986±0.216** using TIC contrast, or 1.032±0.226 using RP, with scatter inflation. Complete eclipse remains viable. Equal-brightness twins remove at most half their combined light and are conditionally disfavored at≈2.3σ; they would need about twice the catalog flux, roughly 0.74 mag brighter. Contrast, color, PRF and stellar-property uncertainties prevent a universal neighbor-EB exclusion.

On the nominal 0.468 Rsun catalog dwarf, a Jupiter-radius object removes only≈4.6% of its own light, giving≈0.07 ppt in the target aperture, far below≈1.5 ppt. An ordinary Jupiter on that specific ordinary dwarf is excluded by the flux budget under these assumptions. The required radius scale is≈0.46-0.48 Rsun. An unequal background stellar eclipse, another unresolved host or erroneous catalog stellar interpretation is a different hypothesis.

The refreshed TESSCut inventory has no genuinely unused sector. SkyMapper 221 resolved-image epochs contain no primary-phase exposure even with 3σ timing allowance; the nearest is 4.95 h from center. DESDR 2 single-epoch SIA returned a parsed empty table after a retry. NSC image/object routes timed out. Those failures do not establish no observations. No public-image forced photometry or upper limit was fabricated.

17 What is excluded, disfavored or viable?

Candidate	Excluded only within stated assumptions	Merely disfavored	Still viable
382098683	Circular, single-lined, edge-on target-host masses above the stated conditional RV profile limit. This is not a universal mass-class exclusion.	Selected massive grazing eccentric fits; separated companions within calibrated template/velocity cells. Exact-apastron examples alone are not decisive.	Target planet; secondary-host planet; faint/coincident/cool companions; wider companion plus planet; some eccentric or 2P stellar/BD blends; noise outside controls.
188830138	Sampled circular bound ordinary main-sequence equal-twin triples under LC+SED assumptions; not all binaries/triples.	Old 3.21σ RV interpretation; circular faint-host planets; strongly eccentric short-period hosts on tidal/population grounds.	Target planet; eccentric secondary-host planet; faint background equal twins at 2P; unequal/eccentric diluted systems; wider companion plus planet.
201247574	Ordinary Jupiter on the nominal 0.468 Rsun 13.7″ neighbor at catalog contrast; prior circular undiluted large-radius tests retain narrow scope.	Equal twins on that faint neighbor at catalog flux; phase-exchangeable noise within fixed controls.	Target planet; unresolved-host planet; nearly total/unequal background eclipse; other diluted/eccentric large companions; unmodeled instrumental/stellar noise.

Shared limitations

Photometric radius or a “planet” scenario label is not a mass measurement. Brown dwarfs and grazing stellar objects cannot all be removed by a small fitted radius under an assumed host. A fine-tuned eccentric geometry is possible without being common. Conversely, a failed selected geometry is not exclusion of its entire class.

Previous resolved-source PRF tests, secondary-search calibration, retrospective early/late comparisons, rotation/harmonic diagnostics and ZTF resolved-neighbor constraints are retained at their original scope. Stellar activity was not established as a coherent dip mechanism, but harmonic amplitude limits and apparently normal Gaia statistics do not prove mass or remove every activity/noise explanation. This continuation does not label retrospective splits as prospective tests.

The restricted blend fits use physical masses in binary Kepler dynamics and finite integrations. They profile local baselines and uncertain stellar scales, but are not a complete population model. They do not assign identical probability to every fitting configuration. SED-derived constraints are used once, and shared-pixel/localization/transit or shared-observation reductions are not multiplied as independent likelihoods.

18 Package FPP versus any-host planet labels

No new FPP draw was needed to answer the convention question. The five executed and report-archived source modules are byte-identical to official TRICERATOPS commit **d20875c27d6761349 cf 032728 bb 4a59 ec 732a84 d**; all 12 saved run records match the main module hash. Thus “locally customized” does not describe a demonstrated difference in these five files. It is not proof of every historical runtime dependency. [7]

Package FPP =1-(TP+PTP+DTP). Those three labels put a planet on the named target. STP puts a planet on an unresolved bound secondary, BTP on an unresolved field star, and NTP on a resolved admitted neighbor. Summing TP+PTP+DTP+STP+BTP+NTP produces an **any-host planet-label weight conditional on the modeled astrophysical scenarios**. It does not include instrumental reliability or independently establish planetary mass.

TIC	Named-target planet weight:4old runs	Any-host planet-label weight:4old runs	Broad existing odds sensitivity
382098683	85.44-87.88%	99.805-99.992%	Target 36.94-98.63%; any 86.73-≈100%
188830138	92.34-95.14%	98.774-99.010%	Target 36.06-99.55%; any 48.32-99.982%
201247574	95.44-98.62%	99.354-99.846%	Target 53.34-99.87%; any 60.71-99.998%

The four old runs are seedA, seedB,1.5× noise and native-bin sensitivity, all sharing observations. The broad envelopes cover the existing 27 simultaneous 0.1/1/10 scenario-odds reweights for each run (324 rows total). These are not credible intervals, new posteriors, or probabilities that real planets exist. Values near 100% simply expose how the enumerated scenario labels divide the modeled probability mass.

All old runs fixed stellar M/R/Teff and set metallicityZ=0. One scalar Gaussian folded-bin error was used. The 1.5× test does not replace a correlated-noise likelihood. Gaia-based field populations can miss undetected stars; no measured contrast curve was supplied. The radius-only “planet” sampler extends 0.5-20 Rearth while binary sampling has a0.1 Msun floor, leaving mass interpretation particularly conditional.

Updating metallicity, SED uncertainty, faint-host populations and aperture-specific likelihoods would require a defensible joint generative model. Adding the new RV/line/PRF scores as independent factors to the old table would double-count some observations or import uncalibrated selection functions. The evidence does not justify that numerical update. Instrumental reliability remains separately reported.

19 Highest-yield remaining observations

Candidate	Smallest useful discriminating observation	Estimated sensitivity and what remains
382098683	One precise H-band transit depth, with optical reference; alternatively calibrated close imaging or a spectrum at opposite 2P quadrature.	Illustrative secondary hosts predict H depth 0.90-1.33 ppt versus 0.43-0.45 ppt for the target. Depth error≈0.15 ppt separates these families at≈3-6σ, before bandpass/red-noise uncertainty. ΔH≈4 mag at 50 mas probes≥6.4 AU; closer hosts remain.
188830138	Two RVs near opposite quadratures at≤1 km/s each; resolve any sufficiently separated faint contaminant with calibrated contrast imaging and in/out photometry.	13 Mjup circular target orbit givesK≈3.06 km/s,≈6.1 km/s difference:≈4.3σ for two 1 km/s measurements. Faint twins contribute 0.14-0.38%, so target RV constancy does not reject them. Their deep eclipse can be tested at≈2% precision if resolved.
201247574	One resolved time series measuring both target and 13.7″ neighbor for≈3.3 h plus≈1 h baseline each side.	Neighbor total-error≤10% of its own flux tests the required≈0.6-1 intrinsic eclipse. Target depth uncertainty≤0.3 ppt tests its shallow dip;≈0.9 ppt per 10 min with 20 in-event and 20 equivalent baseline bins is a photon-noise guide. Include timing/model margin.

For 188, calibrated imaging reachingΔRP≈5.6 mag tests the sampled bound planet hosts only outside its measured inner working angle; ΔT≈7.2 mag reaches the sampled faint twins. Unresolved faint-line detection may require continuum S/N of several hundred even in a favorable multi-line calculation, before mismatch and velocity blending. A normal single-lined low-resolution spectrum is insufficient.

The most useful uncompleted existing-data branches are a successful phase-aware NSC image inventory for 201 and a validated empirical wavelength/LSF recalibration for the LAMOST single exposures. The former is currently access-limited; the latter is calibration-limited. Additional metallicity-dependent binary SED/template work could refine 382's companion map, but the retrieved observations cannot distinguish every coincident/faint host. The exact original discovery selection still lacks a full end-to-end noise calibration.

A future schedule should use the supplied covariance files and allow additional model/systematic timing margin. The frozen ephemerides identify the tests in this report; they are not asserted to have complete future timing uncertainties. No new observing request or message was sent to an observatory.

20 Reproducibility and evidence package

The deliverable directory contains the report, scripts, machine-readable numerical results, diagnostic figures, archive receipts and an input manifest. Tables preserve exact products/epochs, quality flags, assumptions and effects. Source files remain in their original local paths; selected compact dependencies are copied into source_snapshot for transport. Large original pixel cubes are not duplicated into a small result archive: their original paths, hashes and public provenance are retained.

Component	Files / purpose
Root reproducibility	scripts/common.py; inventory.py; protocol_primary.py; primary_conditional.py; matched_controls.py; preprocessing_injections.py; ephemeris_reconstruct.py; reduction_comparison.py; archive fetch/census scripts.
382 spectroscopy	branches/spectra 382: scripts, independent_rv_epochs.csv, full headers, secondary injection maps/null results, circular/eccentric/joint RV grids, approximate host map, inventory and report.
188 spectroscopy / blends	branches/lamost 188: spectral_epochs.csv, new_tests.csv, twin_mass_grid.csv, exact_secondary_host_planets.csv, Roche sensitivity, RV remeasurement windows, source manifests and cached catalog audit.
201 events / neighbor	branches/events 201: event_measurements.csv, new_tests.csv, old_only_ephemeris_covariance.json, background/throughput/timestamp audit, ground image inventory, S28 audit and FPP convention tables.
Archive status	data/archive/*.request.json records requested/final URLs, UTC retrieval, HTTP status, byte count and SHA 256. Malformed/empty/failed/inadequate outcomes remain distinguishable.
All-result index	output/new_tests_all.csv, available_files.csv, README.txt and package_manifest.json identify outputs and dependencies.

Scientific calculations used the local tesshunter Python environment (NumPy/SciPy/pandas/Astropy, batman, numba and relevant source packages). The PDF build uses ReportLab. Versions and reproduction order are in README.txt. Fetch scripts are separate from cached analysis; rerunning an analysis does not silently replace a scientific input with a newer archive product.

Absolute source paths are explicit because those are the actual analyzed inputs. A relocated run must use the snapshot/path map or restore the listed public products. This is a reproducible local result with documented external dependencies, not a claim that the small ZIP contains every multi-GB original pixel input. No branch labels a failed request as a non-detection.

The computation was staged with fixed sample sizes and checkpointed JSON/CSV results. No simulation count was extended to chase favorable significance. The spectral nuisance robustness test and S28 investigation are labeled adaptive diagnostics; they are not extra independent confirmation trials. No astrophysical model fit difference is converted into a calibrated posterior.

21 Sources and product references

[1] MAST TGLC products, sector-list access and caveats
https://archive.stsci.edu/hlsp/tglc

[2] MAST QLP products and current timing notices
https://archive.stsci.edu/hlsp/qlp

[2a] MIT QLP flux columns, flags and TGLC transition
https://tess.mit.edu/qlp/

[3] Astrocut0.11.1 exact cutout timestamp implementation
https://astrocut.readthedocs.io/en/v0.11.1/_modules/astrocut/cube_cut.html

[4] GALAH DR3 release and data access
https://www.galah-survey.org/dr3/

[4a] GALAH DR4 release
https://www.galah-survey.org/dr4/

[5] SDSS DR17 APOGEE spectra
https://www.sdss4.org/dr17/irspec/spectra/

[5a] SDSS DR17 APOGEE radial velocities
https://www.sdss4.org/dr17/irspec/radialvelocities/

[6] LAMOST DR12 release
https://www.lamost.org/dr12/

[6a] LAMOST spectral formats and flags
https://www.lamost.org/dr12/v2.0/doc/lr-data-production-description

[6b] LAMOST single-exposure data and Bai etal.2021
https://dr5.lamost.org/sedr5/

[7] TRICERATOPS audited source commit
https://github.com/stevengiacalone/triceratops/commit/d20875c27d6761349cf032728bb4a59ec732a84d

[7a] TRICERATOPS methodology
https://arxiv.org/abs/2002.00691

[8] LEO-Vetter methodology; no reliability transfer here
https://arxiv.org/abs/2509.10619

[9] ExoFOP documented export functions
https://exofop.ipac.caltech.edu/tess/Introduction_to_ExoFOP_php_functions.php

[10] SkyMapper access and data model
https://skymapper.anu.edu.au/how-to-access/

[11] NOIRLab SIA data access
https://datalab.noirlab.edu/docs/manual/UsingAstroDataLab/DataAccessInterfaces/SimpleImageAccessSIA/SimpleImageAccessSIA.html

[12] Gaia DR3 source/RV definitions
https://gea.esac.esa.int/archive/documentation/GDR3/Gaia_archive/chap_datamodel/sec_dm_main_source_catalogue/ssec_dm_gaia_source.html

[13] Empirical stellar sequence used in conditional maps
https://www.pas.rochester.edu/~emamajek/EEM_dwarf_UBVIJHK_colors_Teff.txt

[14] Roche structure/density sensitivity
https://arxiv.org/abs/1307.4080

[15] IRSA ZTF light-curve API; existing resolved-neighbor analysis retained
https://irsa.ipac.caltech.edu/docs/program_interface/ztf_lightcurve_api.html

Exact observation URLs and identifiers are recorded in the corresponding request manifests rather than abbreviated here. The work uses public TESS/MAST HLSPs, Gaia, GALAH, SDSS/APOGEE, LAMOST and SkyMapper products. Relevant survey acknowledgements are retained in branch documentation. LAMOST data: Guoshoujing Telescope, CSTR 31118.02.LAMOST; cite Bai etal.2021 for single-exposure products.